

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 330

Homework Set 2

Problem 1 (30%) A manufacturer of metal pistons finds that on average, 12% of pistons produced at the facility are rejected because they are either oversize or undersize. Determine the probability that a production batch of 10 pistons will contain **(a, 10%)** exactly 2 rejects, **(b, 10%)** no more than 2 rejects, **(c, 10%)** at least 2 rejects.

Problem 2 (30%) Two fair dice each have 5 sides numbered 1-5. Let X be the sum of the two dice. Determine **(a, 10%)** the probability distribution of X and verify that the sum of all probabilities is equal to 1, **(b, 10%)** the expectation value of X and **(c, 10%)** the standard deviation of X .

Problem 3 (30%) The number of infections X in a hospital each week has been shown to follow a Poisson distribution with mean 3.0 infections per week. Calculate the following probabilities: **(a, 5%)** $P(X = 0)$, **(b, 10%)** $P(X \leq 4)$, **(c, 10%)** $P(X \geq 9)$. **(d, 5%)** Suppose there are 9 infections at the hospital next week. Would you consider this normal? Why?

Problem 4 (10%) A roulette wheel has landed on “red” for each of the past 10 spins. A friend comments that it would be wise to wager on “black” immediately, correctly stating that the odds of the roulette wheel landing on “red” 11 times in a row are below 1%. What is the technical error in this line of thought?